Noise-Robust DTMF Tone Detection Using Goertzel-Based Spectral Analysis and Digital Filtering

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***Abstract*—This paper presents a fully software-based dual-tone multi-frequency (DTMF) receiver that is built from essential Digital Signal Processing topics that are emphasized in Proakis et al, which includes sinusoidal signal modeling, windowing, digital filtering, and discrete Fourier Transform (DFT)-based spectral estimation. The receiver uses a sixth-order Butterworth bandpass front-end followed by either a Goertzel detector or an FFT-based detector. A synthetic but standards-aligned Monte Carlo test-bench is used to evaluate symbol detection under additive white Gaussian noise, low-frequency interference, amplitude twist, and transmitter frequency offsets. Across 6720 trials with 100 ms frames and random $\pm 1.5\%$ tone offsets, the Goertzel detector acheived accuracies of 0.844 at 0 dB, 0.919 at 10 dB, and 0.975 at 30 dB SNR, while matching the performance of a zero-padded FFT baseline and modestly outperforming a plain FFT. Under random frequency offsets of $\pm 1.0\%$ at 0 dB SNR, Goertzel achieved 0.996 accuracy vs 0.956 for the plain FFT. The results show that a targeted Goertzel implementation is a compact and effective DTMF detector, especially when only a small set of frequencies must be tested.**

***Index Terms*—DTMF, Goertzel Algorithm, DFT, FFT, digital filtering, spectral estimation**

I. Introduction

Dual-tone multi-frequency (DTMF) signaling is a classic and still widely recognized application of discrete-time spectral analysis. In the ITU-T specification, each valid keypad symbol is transmitted as the simultaneous sum of one low-group tone and one high-group tone chosen from standardized frequency sets [1], [2]. Since the receiver must verify the presence of exactly one tone from each group, DTMF detection is a natural fit for DFT-based methods and textbook DSP design principles [3].

This projects aims to implement a hardware-free DTMF detector that is intentionally grounded in the standard material of Proakis et al. That is, that the sampled sinusoidal signal models, windowing, IIR filtering, and frequency-selective DFT computation [3]. A Goertzel-based detector is compared against two FFT baselines in a controlled Monte Carlo setting, and the objective is not to claim a novel telephony receiver, rather, to demonstrate a clean, application-driven implementation in which both theory and measured behavior agree.

There are two main contributions here. These are as follows:

1) A reproducible software DTMF detector and receiver.
2) A compact Goertzel detector with an IIR bandpass front-end.
3) A quantitative comparison against plain and zero-padded FFT detectors under noise and frequency-offset stress, and
4) A short frame-length study that exposes the trade-off between spectral selectivity and tolerance to off-nominal tones.

Although DTMF detection is a mature problem, as it remains pedagogically useful since it compresses several major DSP ideas into one implementable application. More specifically, sinusoidal synthesis, frequency-selective filtering, windowed spectral estimation, and decision logic based on group-wise maxima. For this short paper, that combination is more valuable than novelty alone.

II. Background and Signal Model

A. DTMF Signaling

Recommendation Q.23 defines the low-group frequencies as 697, 770, 852, and 941 Hz, and the high-group frequencies as 1209, 1336, 1477, and 1633 Hz [1]. The standard 12-key telephone pad uses the first three high-group frequencies, so a software receiver for the symbols $\{1-9, *, 0, \#\}$ only need to evaluate seven tones. In addition, Q.23 also specifies that each transmitted frequency must remain within $\pm 1.8\%$ of nominal [1]. Moreover, Recommendation Q.24 states that a valid received signal should contain one and only one tone from the low group and one and only one tone from the high group [2].

For a single symbol, the received signal is modeled as follows:

$$\begin{aligned} x[n] =& A_L \sin(2\pi f_L n/f_s + \phi_L) \\ &+ A_H \sin(2\pi f_H n/f_s + \phi_H) + v[n], \end{aligned} \tag{1}$$

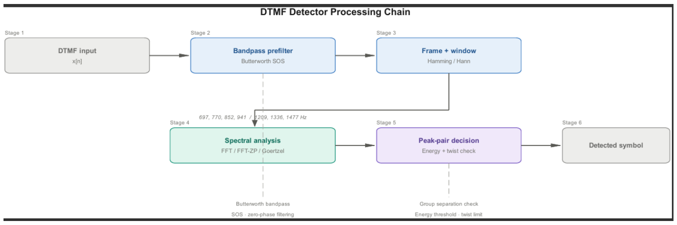

Fig. 1: Software Receiver Architecture used in this project.

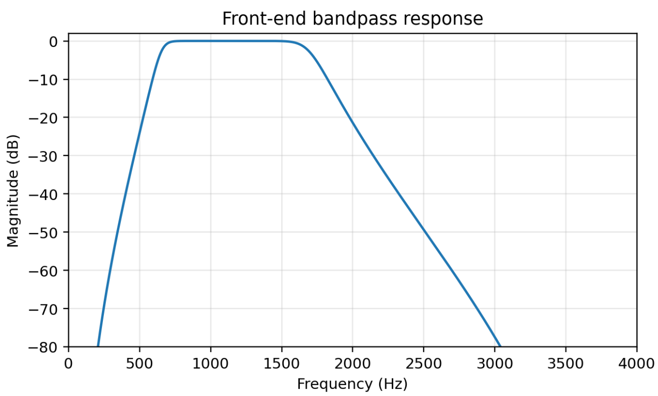

Fig. 2: Bandpass Front-end Magnitude Response.

where A_L, A_H denote and allow amplitude twist between the groups, f_L, f_H denote the DTMF tone pair, and $v[n]$ denotes and captures additive noise and interference.

B. Goertzel Detection

The Goertzel Algorithm efficiently evaluates the selected DFT terms without forming the full transform [4]. For a target digital frequency, given by $\omega_k = 2\pi f_k / f_s$, the recursion is given by the equation:

$$s[n] = x[n] + 2\cos(\omega_k)s[n-1] - s[n-2], \tag{2}$$

with zero initial conditions. After processing a frame of length N, the power at f_k is computed as follows:

$$P_k = s[N-1]^2 + s[N-2]^2 - 2\cos(\omega_k)s[N-1]s[N-2]. \tag{3}$$

Since the keypad subset only requires seven frequencies, the Goertzel is useful here. More specifically, the receiver evaluates only the bins of interest instead of the entire spectrum. In contrast, an FFT-based receiver computes a dense spectrum and then reads out only the few samples that matter.

C. The Design of the Receiver

Figure 1 summarizes the design of the receiver. A synthetic transmitter generates DTMF symbols at $f_s = 8$ kHz and the channel model adds white Gaussian noise, 60 Hz hum, and a small 350 Hz interferer. Before spectral analysis, the signal passes through a sixth-order Butterworth bandpass filter with passband $650-1700$ Hz to suppress out-of-band energy while preserving the DTMF region.

TABLE I: Key implementation parameters

Parameter	Value
Sample rate	8 kHz
Keypad symbols	12 (0–9, *, #)
Target tones	7
Bandpass filter	6th-order Butterworth
Passband	650–1700 Hz
Window	Hamming
Default frame length	100 ms

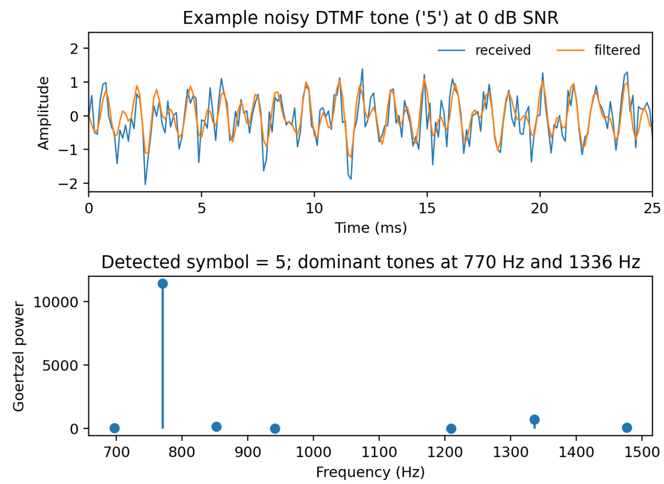

Fig. 3: Example Detection Result for a noisy realization of the symbol ”5”.

This analaysis stage applies a Hamming window and then one of three detectors and these are as follows:

1) Goertzel at the seven nominal DTMF frequencies,
2) a plain FFT followed by nearest-bin selection, and
3) a zero-padded FFT (4096-point) with the same nearest-bin decision rule.

The detector chooses the dominant low-group tone and the dominant high-group tone and maps the pair to the keypad symbol. The released code also exposes an optional twist-consistency check on the dominant pair for use in stricter receiver configurations.

Note that Table I lists the main implementation parameters and the bandpass front-end is intentionally modest as it improves conditioning without making this project depend on highly tuned analog assumptions. Figure 3 shows a representative noisy realization of the symbol ”5” together with the Goertzel response. Even at 0 dB SNR, the correct 770 Hz and 1336 Hz components remain dominant after filtering.

III. Experimental Setup

The evaluation that is conducted in this project is entirely synthetic and fully reproducible. In every Monte Carlo trial, one of the 12 keypad symbols is generated with random amplitude twist uniformly distributed over $[-2, 2]$ dB and random frequency offsets are independently applied to the low-group and high-group tones. The default frame length is 100 ms and for the SNR experiment, the offset bound is fixed

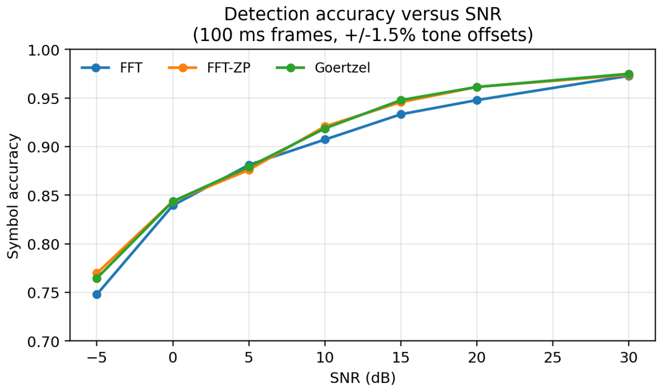

Fig. 4: Detection accuracy versus SNR for 100 ms frames with random $\pm 1.5\%$ tone offsets.

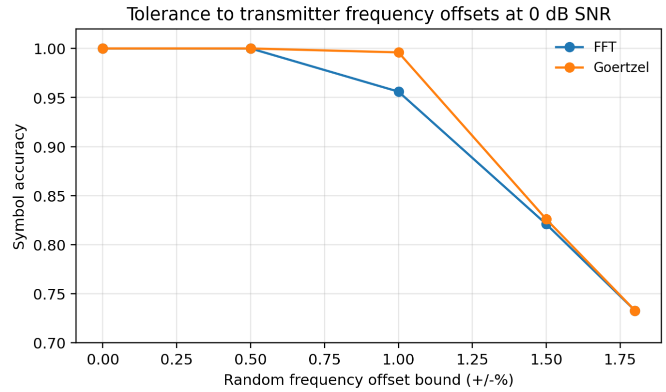

Fig. 5: Tolerance to random transmitter frequency offsets at 0 dB SNR.

at $\pm 1.5\%$ and the SNR is swept from $\pm 1.5\%$ and the SNR is swept from -5 to 30 dB. For the frequency-offset experiment, the SNR is fixed at 0 dB while the offset bound is swept from 0 to $\pm 1.8\%$, which matches the sender tolerance that is mentioned and used in Q.23 [1]. The third experiment varies the analysis-frame duration from 40 ms to 200 ms at 5 dV SNR and $\pm 1.5\%$ tone offsets.

Each operating uses 12 symbols and 80 Monte Carlo realizations per symbol. As a result, the SNR sweep therefore contains 6720 detection trials, and the offset and duration sweeps each contain 4800 trials. These counts are large enough to expose stable trends without increasing the computational complexity of this project.

IV. Results

In Figure 4, there is a comparison between three detectors versus SNR. The figure demonstrates that the Goertzel and the zero-padded FFT are nearly indistinguishable over most of the range, while both outperform the plain FFT slightly at moderate and high SNR. At 10 dB SNR, Goertzel achieves a 0.919 accuracy, as compared with 0.907 for the plain FFT. At 30 dB SNR, all methods approach 0.97-0.98 accuracy, despite the injected tone offsets and group-twist perturbations.

Now, in Figure 5, there is an even more revealing comparison. When the transmitter tones are shifted off nominal, the Goertzel remains more robust than the plain FFT since it evaluates the specified frequencies directly as opposed to relying on the nearest DFT bin. At $\pm 1.0\%$ and 0 dB SNR, Goertzel achieves 0.996 accuracy, while the plain FFT reduces to 0.956.

Finally, in Figure 6, there is an additional practical result. In this figure, with random $\pm 1.5\%$ frequency offsets, the figure reveals that short 40–80 ms frames outperform longer ones at 5 dB SNR. It should be noted this does not contradict Fourier Theory; on the contrary, it reflects it. Longer windows have narrower mainlobes, so a detector that is centered on nominal DTMF tones becomes less forgiving to transmitter offsets. For a simple fixed-frequency receiver, short observation windows can be preferable as a result.

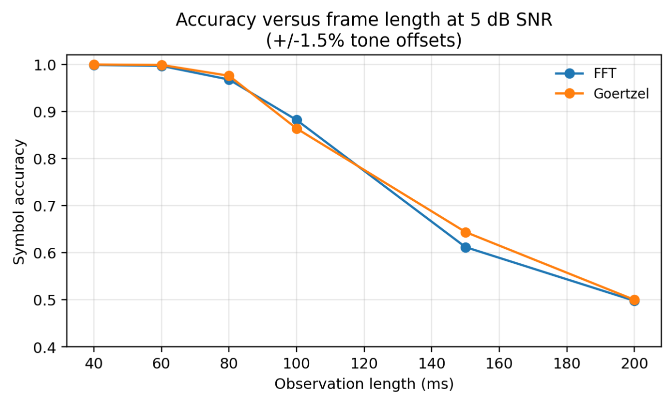

Fig. 6: Effect of frame length on symbol accuracy at 5 dB SNR with random $\pm 1.5\%$ tone offsets.

Finally, Table II summarizes the representative SNR operating points. The zero-padded FFT can match Goertzel closely, however only by computing a denser spectrum than the application strictly requires. In contrast, Goertzel obtains comparable performance while directly testing the seven target tones.

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V. Discussion

There are several textbook DSP concepts that are visible directly in the measured behavior. First, the DTMF detection is fundamentally a sparse spectral-estimation problem; that is, only a few frequencies matter. In addition, that structure demonstrates why Goertzel is an appropriate receiver front-end. Second, the band-pass pre-processing simplifies the problem by restricting the signal to the expected DTMF region before spectral estimation. Third, the offset and frame-length experiments highlight the interaction between observation length, frequency resolution, and mismatch. A longer or more

TABLE II: Representative detection accuracy at selected SNR values

Method	0 dB	10 dB	30 dB
Goertzel	0.844	0.919	0.975
FFT	0.840	0.907	0.973
FFT-ZP	0.844	0.921	0.974

selective is not always beneficial when the source frequencies can deviate from the nominal.

There is also a computational lesson in this project as for the 12-key keypad, only seven spectral tests are required. A Goertzel receiver processes those frequencies directly, whereas the FFT computes a denser spectrum and then discards nearly all of it. And, with a 100 ms frame at 8 kHz, the receiver processes $N = 800$ samples. A Goertzel front-end processes therefore evaluates seven targeted recursions over those 800 samples, while the zero-padded FFT baseline forms a 4096-point spectrum to achieve comparable off-bin accuracy. In this implementation, the Goertzel recursion is executed with a compiled filter primitive as opposed to a Python sample loop, so that the computational argument aligns with the implementation as well as the textbook algorithm. In addition, the project uses a zero-padded FFT that can match the Goertzel's accuracy, however it does so by evaluating many more spectral points than the decision rule ultimately needs.

Another limitation of this project is that the dataset is synthetic and the decision logic is intentionally simple. The FFT-baseline also uses nearest-bin readout, so standard-bin quantization and spectral-leakage effects remain present whenever the DTMF tones fall between FFT bins. More specifically, this is another reason that the Goertzel algorithm edges out the plain FFT under offset stress. The code also exposed an optional twist-consistency check, however a production-quality reciever would still add stricter guard times and more elaborate rejection logic for invalid symbols. Nevertheless, for this project centered around Proakis-style DSP implementation, the current receiver is already complete as it is reproducible, application-based, and grounded in standard DFT and filter-design tools.

VI. Conclusion

In conclusion, we present a compact DTMF detector that was built entirely in software from classical DSP components. The implementation in this project combines a Butterworth front-end with either Goertzel or FFT-based spectral estimation and was evaluated under noise, interference, as well as standards-aligned frequency offsets. The Goertzel algorithm matched the accuracy of a zero-padded FFT and outperformed a plain FFT under moderate frequency offsets while requiring evaluation of only the seven frequencies needed by the 12-key keypad. The code that was used in this project also ensures reproducibility as all the figures and tables can be reimplementedand and also could be extended with threshold logic, guard-time tests, or alternative filter desgins.

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